

MATH 136

Lagrange Multipliers

by

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Lagrange Multiplier's

Suppose the curve $g(x, y) = k$ is made out of wire embedded in a metal plate and that the temperature of the flat plate at the point (x, y) is $f(x, y)$. How do we find the point on the wire which is hottest?

Generally, we often wish to maximize a function $f(x, y)$ subject to the constraint that $g(x, y) = k$.

If we can solve $g(x, y) = k$ to obtain $y = \varphi(x)$ then we can do the problem by maximizing the function $f(x, \varphi(x))$ of one variable using high-school methods. However it is generally impossible to find the explicit function φ .

Lagrange found a way to do this kind of problem without solving $g(x, y) = k$ for y .

Technique: Look at the function

$$h(x, y) = f(x, y) + \lambda g(x, y)$$

where λ is called the Lagrange multiplier.

The critical points for f subject to $g = k$ are solutions (x, y) of the three equations

$$\frac{\partial h}{\partial x} = 0, \quad \frac{\partial h}{\partial y} = 0, \quad g(x, y) = k \quad \text{Lagrange (2)}$$

in the three variables x, y, λ .

Example Find the points on the curve

$$17x^2 + 12xy + 8y^2 = 100$$

closest & farthest from the origin.

Solution It is easier to use $f(x, y) = x^2 + y^2$ rather than $\sqrt{x^2 + y^2}$ (= distance from (x, y) to $(0, 0)$); the max. & min. will be the same. So we must find critical points of $f(x, y)$ constrained by $g(x, y) = 17x^2 + 12xy + 8y^2 = 100$. Introduce the Lagrange multiplier λ :

$$h(x, y) = x^2 + y^2 + \lambda(17x^2 + 12xy + 8y^2).$$

We must solve the three equations

$$\left. \begin{aligned} \frac{\partial h}{\partial x} &= 2x + 34\lambda x + 12\lambda y = 0 \\ \frac{\partial h}{\partial y} &= 2y + 12\lambda x + 16\lambda y = 0 \end{aligned} \right| 17x^2 + 12xy + 8y^2 = 100$$

for x, y, λ . The left equations give

$$\left. \begin{aligned} x + (17x + 6y)\lambda &= 0 \\ y + (6x + 8y)\lambda &= 0 \end{aligned} \right\} \Rightarrow (6x + 8y)x = (17x + 6y)y$$

This gives $2x^2 - 3xy - 2y^2 = 0$. Multiply by 4 and add to the constraint equation:

$$25x^2 = 100, \quad x^2 = 4, \quad x = \pm 2$$

Putting these values of x in the constraint equation gives two quadratic equations

for y : $y^2 + 3y - 4 = 0$, $y^2 - 3y - 4 = 0$. Lagrange (3)

This gives four critical points

$$(2, 1), (-2, -1), (2, -4), (-2, 4).$$

(The first two are the ends of the minor axis of the ellipse & the second two are the ends of the major axis.)

$$f(2, 1) = 4 + 1 = 5 = f(-2, -1)$$

$$f(2, -4) = 4 + 16 = 20 = f(-2, 4)$$

The shortest distance from the origin to the ellipse is $\sqrt{5}$ and the farthest is $\sqrt{20}$.

The same technique works with maximizing $f(x, y, z)$ subject to a constraint $g(x, y, z) = k$:
look at

$$h(x, y, z) = f(x, y, z) + \lambda g(x, y, z)$$

and solve the four equations

$$\frac{\partial h}{\partial x} = \frac{\partial h}{\partial y} = \frac{\partial h}{\partial z} = 0, \quad g(x, y, z) = k$$

for λ, x, y, z .

Example Find the shortest distance from the origin to the surface $xyz^2 = 2$.

Solution We need to minimize

$$f(x, y, z) = x^2 + y^2 + z^2$$

subject to $xyz^2 = 2$. Look at

$$h(x, y, z) = x^2 + y^2 + z^2 + \lambda xyz^2.$$

$$\left. \begin{aligned} \frac{\partial h}{\partial x} &= 2x + \lambda y z^2 = 0 \\ \frac{\partial h}{\partial y} &= 2y + \lambda x z^2 = 0 \\ \frac{\partial h}{\partial z} &= 2z + 2\lambda xy z = 0 \end{aligned} \right| \quad x y z^2 = 2.$$

From the right-hand equation we see that x, y, z are all non-zero. From the left equations we get

$$\lambda = -\frac{2x}{y z^2} = -\frac{2y}{x z^2} = -\frac{1}{xy}. \text{ This yields}$$

$2x^2 = z^2 = 2y^2$. So $x = \pm y$ and, from the constraint, $x(\pm x)2x^2 = 2$; so $\pm x^4 = 1$.

But $x^4 = -1$ is impossible; so $x^4 = 1$. So $x = \pm 1$. From the constraint equation we also see that x, y have the same sign; so $x = y$.

So the critical points are

$$(1, 1, \sqrt{2}), (1, 1, -\sqrt{2}), (-1, -1, \sqrt{2}), (-1, -1, -\sqrt{2}).$$

Notice that $f(x, y, z) = 1 + 1 + 2 = 4$ is the same for all four points (x, y, z) . So these are the closest points and the shortest distance is $\sqrt{4} = 2$.